Ron Medina

(+63) 976 459 2991 • ron.mdn@gmail.com
linkedin.com/in/ron-medina-5a411218a/ • particle1331

Machine Learning Engineer with a mathematics background. I build and ship ML and deep learning systems that optimize operations and surface useful insight — with attention to the infrastructure and engineering that keeps them running in production.

Employment History

Machine Learning Engineer | Ubiquity — Aug 2022 – Present

- R&D / creating POCs and MVPs of AI products applicable to client domain
- Perform evaluation, cost-estimation, deployment strategies based on infrastructure constraints and business goals
- Developed and deployed transcription service which transcribed 1.2M+ production calls
- Reduced cost by 6X and increased accuracy of speech recognition system by +10%
- Developed and designed a fault tolerant distributed offline task queue service to scale speech recognition
- Helped develop the backend application for searching and filtering transcriptions
- Developed AI services for downstream processing, modeling, and analytics of call transcripts
- Extended existing open-source annotation tool for human data labeling to serve our internal use-case
- Developed a Transformer-based semantic search engine that runs performant on CPU
- Helped develop a masking model for images (screenshots) containing sensitive data using Tesseract
- Within 3 months after hiring, presented a POC for extending call recorder system from mono to stereo recording using socket programming and Avaya APIs which became the basis for a major project for the Telecommunications team.

Junior Data Scientist | Sitel — June 2021 – Aug 2022

- Works directly under the Director of District Operations Quality Management
- Design of KPIs and metrics for various business processes
- Creation and deployment of Power BI dashboards
- Creation and deployment of APIs for integrating ML algorithms with existing products
- Ensure integrity and accuracy of reports

Machine Learning Engineer | BrewedLogic, Inc. — Mar 2020 – June 2021

- Preprocessing, analysis, and feature engineering of datasets (1M-5M) for model training, selection, and evaluation
- Research and implementation of novel techniques and architectures for RecSys and fraud detection
- Developed RecSys used by Crisp restaurant platform and deployed on 26 franchises each with multiple stores in the US
- The collaborative filtering based model was written using NumPy, Scikit-Learn, and Pandas, and served in a Django server. Deployment of food RecSys increased average ticket count from 3.38 to 4.86 and average ticket value from $11.49 to $14.21 after the first two months in production.
- Worked with a senior data scientist on customer segmentation and sales forecasting
- Developed fraud detection system for fraudulent VoIP transactions which drastically improved previous rule-based approaches
- Maintenance, testing, and monitoring of deployed models, and service migration to FastAPI

Bootcamp Associate | Eskwelabs — Oct 2019 – Jan 2020

- Developed curriculum materials on machine learning algorithms
- Facilitated live Hackathons
- Presented a talk at the National Youth Congress held at UP Diliman School of Economics, Nov 2019, on the topic of artificial intelligence and deep learning

Data Analyst | Tita's Groceria — June 2017 – June 2019

- Worked part-time at an E-commerce shop with hundreds of daily transactions.
- Performed basic statistical analysis of customer data including clustering based on RFM criteria.
- Analysis of frequently bought together items using graphs, Markov chains, and correlations.
- Created reports about product ROI rate.
- Helped grew follower count from 30,000 to 100,000+.

Skills

Data Analysis

- Data visualization, data wrangling, and EDA using Pandas, seaborn, matplotlib, and NumPy.
- SQL, probability modelling, statistics, clustering

Machine Learning

- Deep neural networks in TensorFlow and PyTorch
- Machine learning models in scikit-learn
- Recommender systems, anomaly detection / imbalanced learning
- Weak supervision for training noise-aware models
- Gradient Boosting (Catboost, XGBoost, LightGBM), ensembling/stacking

Model Deployment and MLOps

- Development and deployment of prediction APIs (Django, FastAPI, Flask)
- Automation: Gitlab CI/CD, pdm, GitHub Actions, tox, Makefiles
- Task management with Celery
- MLflow experiment tracking, system metrics, evaluation, and model management
- Production quality Python packages with pdm
- Version control using git
- Unit testing with pytest, differential, and regression testing
- Containerization with Docker
- AWS Lambda, SQS and RabbitMQ, S3, RDS, EC2 / Auto Scaling groups
- PostgreSQL, MySQL, and Redis

Algorithms and Data Structures

- Gold level in Problem Solving and Python @ Hackerrank

Writing

- Author of OK Transformer — a collection of notebooks and articles on deep learning, ML engineering, and MLOps. Auto build/deploy via GitHub Actions + tox. Featured in the Gallery of Jupyter Books.
- Contributed to the mathematics appendix of Dive into Deep Learning; acknowledged as a contributor.

Education

University of the Philippines - Diliman **06/13 – 12/18 (courses), 09/23 – 01/24 (thesis)**
Bachelor of Science
Major in Mathematics

Awards: University Scholar, 2nd Semester 2013-2014. GWA: 1.23
Thesis: *An Introduction to Finite Frames and a QR Factorization Approach for Constructing MB Frames*
Relevant courses: Introduction to Computer Science (Python), Numerical Analysis, General Relativity (Graduate level), Linear Algebra, Advanced Calculus

Eskwelabs **July 2019 – Oct 2019**
Data Science Bootcamp
Attended a 10-week bootcamp which included 160 hours of in-class learning in addition to coursework. At the end of the bootcamp, I presented my capstone project about modeling nonlinear chaotic systems using neural networks implemented in TensorFlow 1.x to industry leaders in Makati City, Philippines.